

G10 Shared Hub and Caption Head

A consolidated explanation of hub composition, target overlap, held-out encoders, effective rank, and text-to-image reversal

Dataset: 9,533 paired items: 8,533 training rows and 1,000 strictly held-out evaluation rows.

This document consolidates the discussion and corrects two important labels: the 11.2-point result is a full roster effect, while the matching-target interaction is approximately 3.0 points on average.

Core idea

Multiple image and text encoders are mapped into one 512-dimensional coordinate system. A reusable linear head then translates hub coordinates into a chosen target embedding space, such as BGE.

raw encoder vector -> encoder entry map -> 512-d Hub -> reusable head -> target embedding

1. Encoder roles in G10

Encoder	Role
DINOv2-small	Hub member; fixed source used to train the reusable caption head
DINOv2-base	Hub member and transfer evaluation
DINOv2-large	Hub member and transfer evaluation
BGE	Text hub variant and caption-head target
SBERT	Text hub variant and caption-head target
BERT	Text hub variant only
GPT-2	Text hub variant only
SigLIP	Held-out image architecture; new entry map only
ConvNeXt	Held-out image architecture; new entry map only

The later seven-space G1-style roster is three DINOv2 image spaces plus BGE, SBERT, BERT, and GPT-2. ConvNeXt and SigLIP do not construct that hub.

2. Best-to-worst crossed BGE/SBERT results

Rank	Hub	Head target	% of native	Gain vs images-only
1	Images + SBERT	SBERT	97.9%	+9.8 pt
2	Images + BGE	SBERT	95.9%	+7.9 pt
3	Images + BGE	BGE	95.3%	+9.8 pt
4	Images + SBERT	BGE	91.3%	+5.8 pt
5	Images only	SBERT	88.1%	baseline
6	Images only	BGE	85.5%	baseline

3. What the matching effect means

The measured BGE-hub to BGE-target result remains **95.3%**. Nothing is subtracted from that performance. Subtraction is used only to decompose why it improved over the image-only hub.

Comparison	Calculation	Interpretation
BGE hub -> BGE target	$95.3 - 85.5 = +9.8$	Total benefit over images-only
SBERT hub -> BGE target	$91.3 - 85.5 = +5.8$	A nonmatching text space also helps
BGE rather than SBERT -> BGE	$95.3 - 91.3 = +4.0$	BGE-specific matching advantage
SBERT rather than BGE -> SBERT	$97.9 - 95.9 = +2.0$	SBERT-specific matching advantage
Average matching advantage	$(4.0 + 2.0) / 2 = 3.0$	Crossed-control estimate

Approximate decomposition

6.8 points generic text-space benefit + 3.0 points matching-target benefit = 9.8 points total matching gain

The 3.0-point value is an average interaction estimate across only BGE and SBERT. It should not be presented as a universal constant.

4. Target-informed is not train/test leakage

Images + BGE -> BGE is target-informed because BGE helps construct the hub and is also the head target. **Images + BGE -> SBERT** is more target-independent because SBERT did not construct that hub.

The target-informed protocol is still valid: all fitting uses the 8,533 training rows and evaluation uses the untouched 1,000 rows. The dependency must be declared, but it is not conventional train/test leakage.

5. Hub text composition matters

Hub text composition	BGE-target mean (% native)
All four text spaces	96.7%
BGE only	95.3%
SBERT only	91.3%
GPT-2 only	90.3%
BERT only	89.2%
No text space	85.5%

Two effects must not be confused:

Single-member identity spread: $95.3 - 89.2 = 6.1$ points.

Full roster spread: $96.7 - 85.5 = 11.2$ points (none versus all four).

Therefore, the earlier inference that text-member choice should barely matter was falsified. The 11.2-point value is not purely a choice among single members; it also changes the number of text spaces.

6. Low global reach can still be task-critical

E.6 asked how much of the complete hub a text encoder can reconstruct. Its low R-squared values (roughly 0.02-0.05) indicate low *global* linear reach. G10 asked whether text participation changes downstream transfer. These are different questions.

E.6 reachability	G10 sensitivity
How much of the whole hub can text reconstruct?	How much does text membership change retrieval?
Global recoverable variance	Task-specific usefulness
About 2-5% reach	6.1-point identity and 11.2-point roster effects

A small semantic subspace may explain little total variance while containing exactly the directions needed for caption retrieval. Small variance does not mean small importance.

7. Effective rank

Ordinary rank counts every nonzero singular value. Effective rank estimates how many dimensions meaningfully carry structure. A nominally 512-dimensional hub may behave as though far fewer directions matter.

Entropy effective rank: $p_i = \sigma_i / \sum(\sigma_j)$, $S = -\sum(p_i \log p_i)$, $r_{\text{eff}} = \exp(S)$

Participation ratio: $q_i = \sigma_i^2 / \sum(\sigma_j^2)$, $r_{\text{PR}} = 1 / \sum(q_i^2)$

Spectrum	Interpretation
All 512 singular values similarly strong	Effective rank near 512
About 100 meaningful directions	Effective rank near 100
One direction dominates	Effective rank near 1

A text-derived hub representation can have low effective rank yet remain excellent for caption prediction if its limited directions are semantically aligned with the task.

8. Why DINOv2-small trains the caption head

DINOv2-small is not physically part of the final head. Its mapped training coordinates are the inputs used to fit one 512-to-1024 matrix:

$$H_{\text{small,train}} = X_{\text{small,train}} W_{\text{small}}$$

$$H_{\text{small,train}} W_{\text{head}} \text{ approximately equals } T_{\text{BGE,train}}$$

DINO-small is the fixed source encoder, not a teacher in the knowledge-distillation sense. DINO-base or DINO-large could have been selected instead; G10 does not establish that small is the best source.

9. What happens on the 1,000 held-out inputs

The held-out inputs never fit the hub, entry maps, or head. For a ConvNeXt evaluation row, the path is:

ConvNeXt test vector -> W_ConvNeXt -> Hub -> frozen W_head -> predicted BGE vector

DINOv2-small is not executed in this route. It was used earlier only to fit the reusable head on training rows.

Stage	Rows	What is fitted
Hub construction	8,533 train	Whitened shared basis
Entry maps	8,533 train	One ridge map per encoder
Caption head	8,533 train	DINO-small hub -> BGE
ConvNeXt/SigLIP maps	8,533 train	New encoder -> existing hub
Final retrieval	1,000 held out	Nothing; evaluation only

The source encoder is deliberately excluded from G10's transfer mean. The mean covers DINO-base, DINO-large, SigLIP, and ConvNeXt, so it measures reuse beyond the head-training source.

10. Why ConvNeXt and SigLIP are meaningful

They are architecture-held-out, not data-free. Each receives a newly fitted linear entry map, but the hub and caption head remain frozen. This is new-encoder adaptation, not zero-shot adaptation.

With the all-four-text hub, ConvNeXt reached 99.3% of its own direct ConvNeXt-to-BGE native baseline, showing that the shared head transfers almost all of the direct baseline performance.

11. Text query to matching image: two designs

Design A - existing system

The existing caption head already supports text-to-image retrieval by moving gallery images into BGE space:

gallery image -> image encoder -> Hub -> BGE head -> predicted BGE vector

text query -> BGE query vector; compare it with every predicted gallery vector using cosine similarity.

This retrieves images from text, but does not predict an image embedding from text.

Design B - true reverse-direction head

To predict an image embedding directly, train a Hub-to-image head. For a DINO-base target:

$H_{BGE,train} W_{image}$ approximately equals $X_{DINO-base,train}$

text -> BGE -> W_{BGE} -> Hub -> W_{image} -> predicted DINO-base vector -> image retrieval

Direction	Source entering hub	Head target	Retrieval space
Image -> text	DINO / ConvNeXt / SigLIP	BGE	BGE text space
Text -> image	BGE / SBERT / BERT / GPT-2	DINO / ConvNeXt / SigLIP	Image-encoder space

12. Reverse transfer and experimental purity

To test cross-text transfer, train the image head using BGE hub coordinates, freeze it, fit only an SBERT entry map, and evaluate held-out SBERT captions through the same head:

SBERT -> W_SBERT -> Hub -> frozen BGE-trained image head -> predicted DINO-base vector

Success would be the reverse counterpart of training the caption head from DINO-small and reusing it with ConvNeXt or SigLIP.

Reverse hub setup	Image-head target	Interpretation
Hub includes DINO-base	DINO-base	Target-informed
Hub excludes DINO-base	DINO-base	More target-independent
Hub includes DINO-small	DINO-base	Cross-image-target transfer

A particularly strong reverse test is BGE input -> hub without DINO-base -> DINO-base prediction. The target image space did not help construct the hub.

13. Final conclusions

1. An image-only hub works, but adding text substantially improves text prediction.
2. Matching the hub text member to the target adds approximately 3.0 points on average across BGE and SBERT.
3. The matching setup is target-informed, not train/test leakage.
4. Text composition matters even though text has low global reach into the hub.
5. A single DINO-small-trained head transfers to other image architectures after fitting only their entry maps.
6. The current system already retrieves images from text in BGE space; a true reverse experiment requires a Hub-to-image head.